NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

CS5224 CLOUD COMPUTING · FINAL PRESENTATION · DYLAN CHAN · XIN WEI CHONG · KELVIN LOW · JOSHUA LUM · BRANDON TAN

$10k+

to test one ad on a brain scanner

fMRI study	**$10 – 15k**	2–4 weeks
NeuroLens	**$1.00**	≈ 7 min

Ten cuts by lunch.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



What already exists: measured, filmed or guessed.

Brain scans (fMRI, EEG) research labs, iMotions suites	$10,000 – $15,000	2–4 weeks
Facial coding Realeyes, iMotions webcam studies	$850 – $4,000	3–5 days
AI attention maps Neurons AI, Attention Insight	$500 – $1,500 / mo	sub-minute
NeuroLens predicted brain response	$1.00	≈ 7 min

OUR DIFFERENCE

A brain response, not a gaze or a face

Per second, with a no-sound pass

Compare any two cuts for $1.00

Under the hood: TRIBE v2.

TRAINED ON REAL BRAINS

1,000+ hours of fMRI

From 720 people, on naturalistic video, audio and language.

THREE SENSES IN

Frames, sound, words

V-JEPA 2 watches, Wav2Vec-BERT listens, WhisperX and LLaMA 3.2 read the speech.

BRAIN OUT

20,484 points a second

Predicted on the cortex for new videos and new people, several-fold better than linear models.

Pay as you go. Credit, not cards.

WHO AND HOW MUCH

$1.00 per started 30 s

For independent creators and small agencies picking one of many AI-made cuts. New users get $10 of credit: ten videos.

WHY PREPAID CREDIT

Card fees are fixed

About 2.9% + $0.30 a charge: a third of a $1.00 sale, but only 4% of a $20 pack.

Course version: an administrator grants credit; Stripe Checkout belongs to the production design. TRIBE v2 is CC BY-NC, so a paid launch needs a licence.

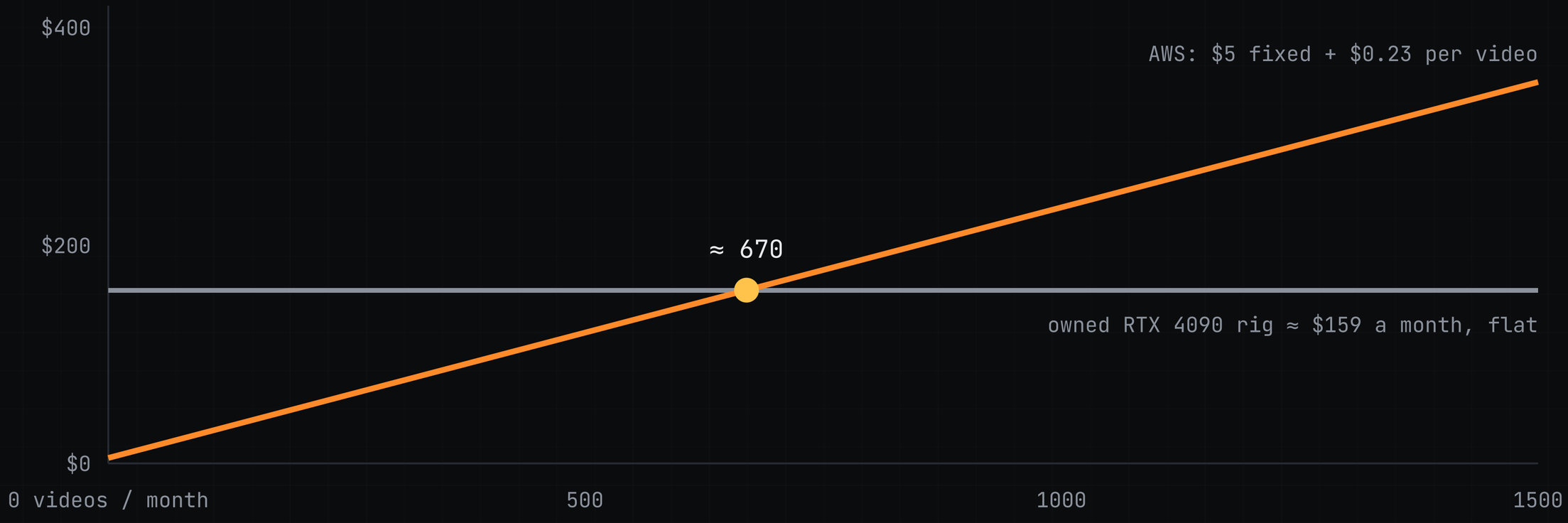
What one analysis costs us.

$0.14–0.23

our AWS cost per 30 s video, against a $1.00 price

GPU: 393 s at $1.22/h (L4)	$0.13
Start-up + idle, per session	7.5 + 7 min
Cost, 3 videos per session	$0.23
Cost, 30 videos per session	$0.14
Fixed cost per month	$5
Gross margin	**77 – 86%**

Cloud is cheaper below about 670 videos a month.



One upload in.
Seven signals out.

Your video

MP4, MOV or WebM, up to two minutes

TRIBE v2

Predicts the brain's response every second

Seven signals

Seeing: faces, bodies, scenes, visual · Hearing: sound · Understanding: speech, social

Three outputs

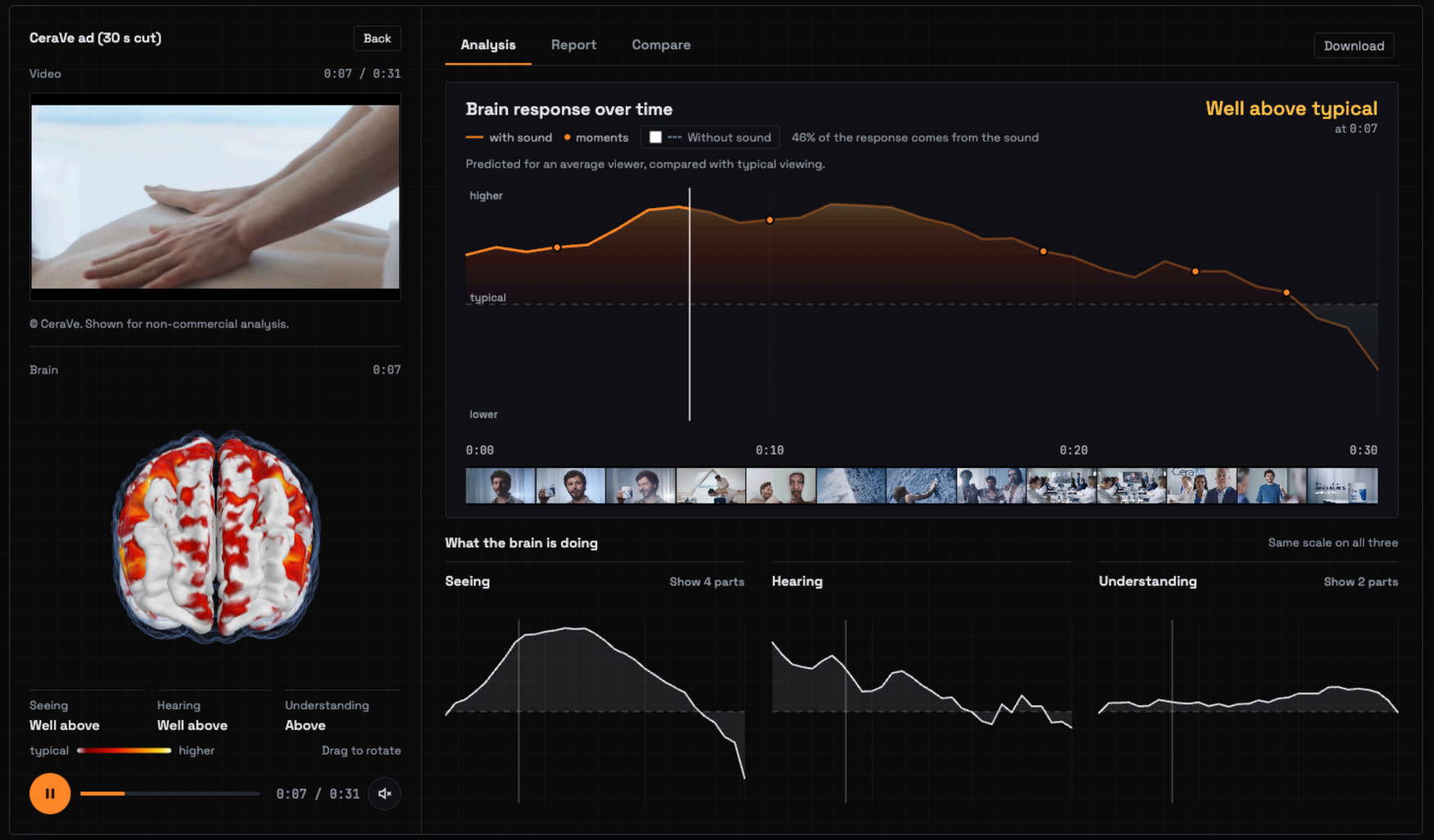
Timeline · 3D brain · AI report

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

LIVE Live demo: sign in, upload, explore



1. Upload: price first, then confirm
2. Sample result: peak, dip, no-sound toggle
3. AI report and Compare tabs
4. A bad file is rejected, nothing charged

CLICK THE PICTURE TO OPEN THE LIVE SITE · DF6GIRQYYKYFA.CLOUDFRONT.NET

We pay for GPU minutes only when someone is waiting.



Check first

Length, size and credit are verified before any GPU wakes.

Scale to zero

Off after 7 idle minutes: about $5 a month when idle.

Refund by default

Credit is charged or refunded exactly once. A failed job costs nothing.

How do we know it works?

THE MODEL

Validated by Meta

Accurate on new stimuli, tasks and people (d'Ascoli et al., 2026). We did not re-test it against real scans.

OUR PIPELINE

Same output everywhere

RTX 4090 vs AWS: 11 series, largest difference 0.005, correlation 1.000. 77 of 80 refund tests passed.

OUR USERS

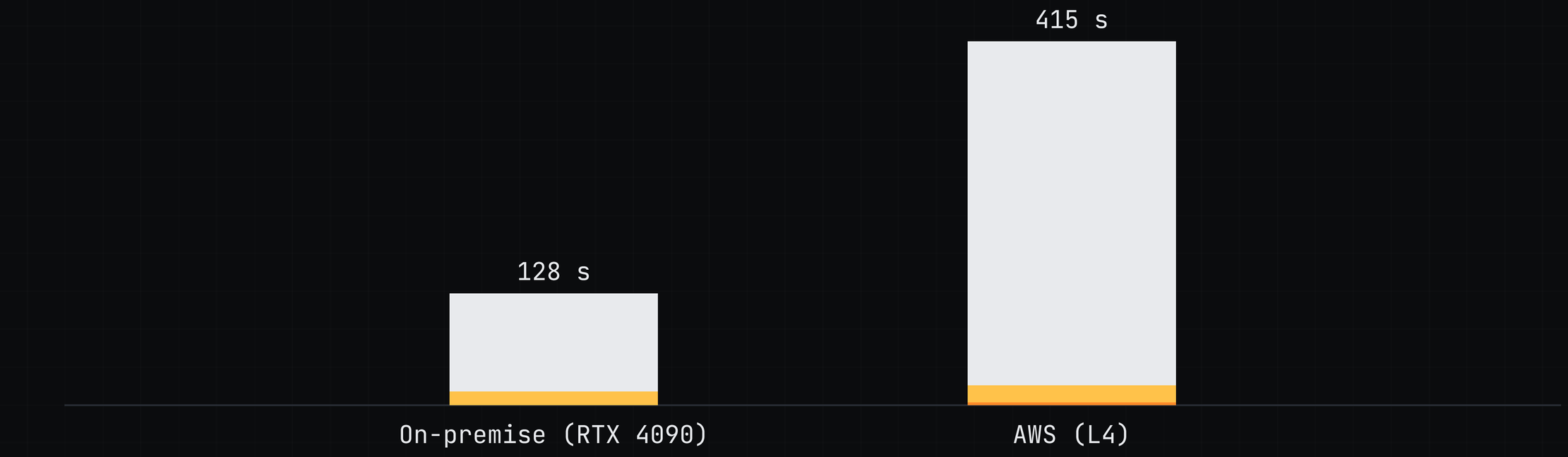
Usability and value

SUS and a short booth survey: see Usability.

Not validated: that a higher response means more views or sales.

Where does the time go in one job?

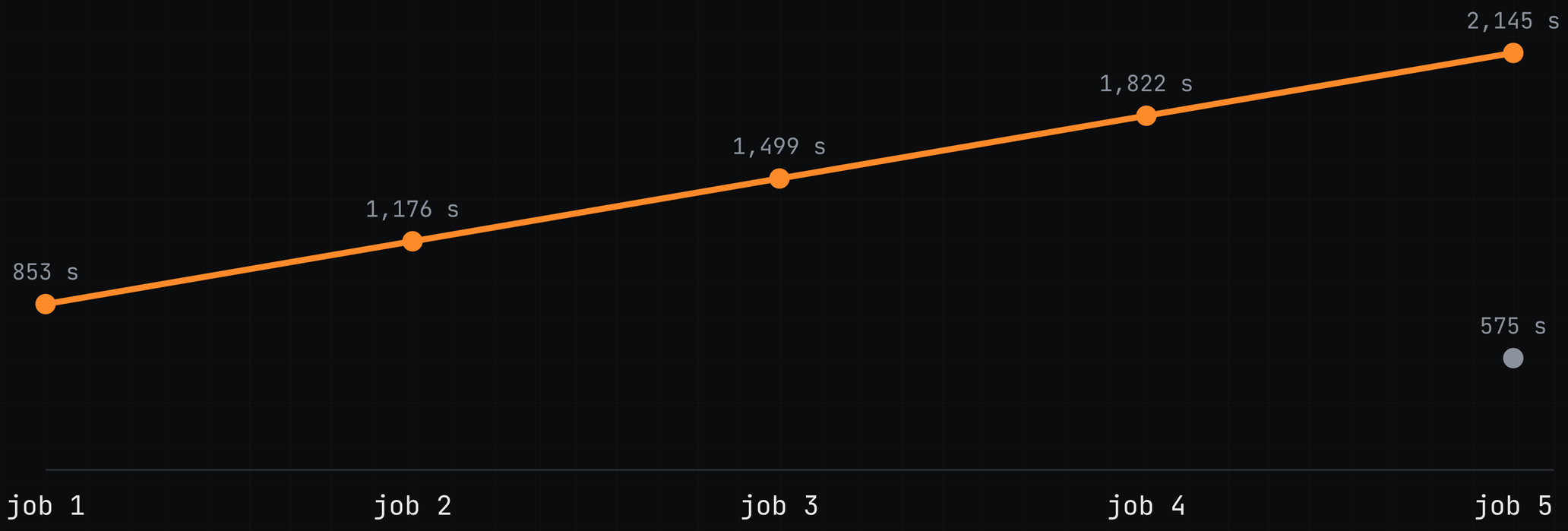
Same code, same 30 s clip, on our own RTX 4090 and on AWS.



GPU TIME PER VIDEO-SECOND: 3.71 S (RTX 4090) VS 10.7 S (A10G) · FIRST VIDEO AFTER 7+ MIN IDLE: 779 S ON AWS

Five people at once.

Five uploads at once. The endpoint is set to one GPU, so jobs run one after another.



SECONDS UNTIL EACH JOB FINISHES. NO JOB WAS DROPPED. THE AWS RUN GOT AN A10G, AND ITS FIRST JOB APPEARS TO HAVE RESTARTED ONCE.

Can people use it?

Booth visitors tried a sample or uploaded a clip, then answered the 10-question SUS.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS VERY GOOD

[X]% SAW SOMETHING THEY HAD NOT NOTICED
[Y]% WOULD PAY $1.00 FOR A 30 S VIDEO

What it claims, and what it doesn’t.

WHAT IT IS

- An average viewer’s predicted brain response
- One scale for every video
- A virtual pre-test for editing

WHAT IT ISN'T

- A prediction of views or sales
- Instant: first video after a quiet spell ≈ 6 min
- Statistically proven (small test)

ETHICS

- CC BY-NC: no commercial use without a licence
- No real person is scanned or profiled
- For checking edits, not manipulating people

Where next: scaling up.

Spot did not deliver

All 26 Spot launches failed for lack of capacity. Spot averaged only 1–9% below on-demand.

A smaller GPU fits

The model peaked at 19.9 GB, so a 24 GB L4 or A10G is enough.

The GPU sets the speed

119 s clip: 491 s on an L40S, 1,540 s on an L4.

Production design (EC2 GPU pool behind a load balancer): $0.10 – $0.20 a video, about $70 a month more fixed cost. Cheaper than SageMaker on the same L40S above about 480–1,440 videos a month. Not built: estimates.

A $10,000 brain study, in 7 minutes for $1.

Live today: sign-in, upload, prediction, 3D brain, AI report, comparison, refunds.

Next

Licence

TRIBE v2 is CC BY-NC: a paid launch needs a licence.

Next

Scale

A GPU pool that grows with the queue: cheaper above about 1,000 videos a month.

Next

Polish

Fix what our usability study flagged.

Thank you.
Questions?

df6girqyykyfa.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS